Weighted grading for the k = 5 Joux–Vitse Edwards test: a loss

2026-11-06 · probe `weighted_probe` (`src/cryptanalysis/jv_quintic_edwards.rs`) · p = 271, seed 2, three random residuals

Question. Does the weighted grading that cut the D_4 route's solve also cut the 2-torsion Edwards route's per-residual decision cost?

Answer. No. It decides only at weighted degree 37, on a 28,981 × 26,618 Macaulay matrix costing $3.9 \cdot 10^{10}$ F_p multiplications. That is about 670× more than `f4_fp`'s $5.85 \cdot 10^{7}$. The unweighted Macaulay solve decides at degree 9 for $7.0 \cdot 10^{7}$. Every figure is an unverified assertion.

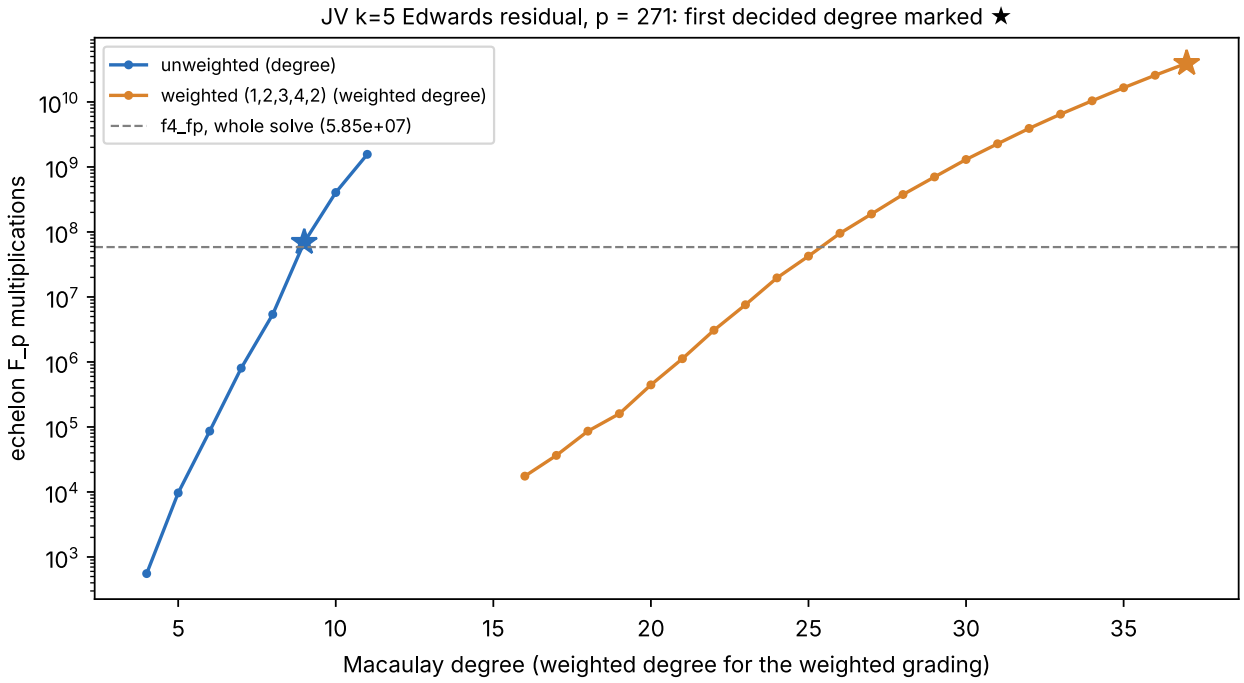

Figure 1: Echelon multiplications against Macaulay degree; ★ marks the first decided degree. The dashed line is f4_fp's whole solve.

solver	decided at	matrix	F_p mults
f4_fp grevlex	inconsistent	≤ 720 × 768	$5.85 \cdot 10^{7}$
Macaulay unweighted	degree 9	1,802 × 2,002	$7.0 \cdot 10^{7}$
Macaulay weighted (1,2,3,4,2)	weighted 37	28,981 × 26,618	$3.9 \cdot 10^{10}$

Why. The symmetrised equations are bounded by total degree 4 in the e_i, so their weighted degree ranges up to 16. The weighted grading fits them worse than the plain one.

Precomputing the target-independent part. Every trial lies in one fixed 105-monomial space, and its five combinations are generic. Trace replay already exploits the fixed shape, and no smaller system follows from it. Not built.

Not claimed. No change to C″, the leaderboard or benchmarks. One prime. A local probe.